Kou (2002): A Jump-Diffusion Model for Option Pricing

Introduction

The Black–Scholes model assumes log-normally distributed asset returns, but two well-documented empirical failures motivate Kou (2002). First, real return distributions exhibit a *leptokurtic* shape: a higher peak and heavier, asymmetric tails than the normal distribution. Second, implied volatilities backed out from market prices are not flat across strikes but form a convex *volatility smile*, meaning Black–Scholes systematically underprices options with extreme strikes. Kou proposes a double exponential jump-diffusion model that addresses both problems while retaining analytical tractability for a wide class of derivatives.

The Double Exponential Jump-Diffusion Model

The stock price is governed by:

$$\frac{dS(t)}{S(t^-)} = \mu\, dt + \sigma\, dW(t) + d\left(\sum_{i=1}^{N(t)} (V_i - 1)\right) \tag{1}$$

where $W(t)$ is a standard Brownian motion, $N(t)$ a Poisson process with intensity λ, and $\{V_i\}$ i.i.d. non-negative jump multipliers independent of $W(t)$ and $N(t)$. The log-jump size $Y = \log V$ follows an asymmetric double exponential distribution:

$$f_Y(y) = p\,\eta_1 e^{-\eta_1 y}\mathbf{1}_{\{y\geq 0\}} + q\,\eta_2 e^{\eta_2 y}\mathbf{1}_{\{y<0\}}, \qquad \eta_1 > 1,\ \eta_2 > 0 \tag{2}$$

where $p+q = 1$ are the probabilities of upward and downward jumps, with mean jump sizes $1/\eta_1$ and $1/\eta_2$ respectively. The constraint $\eta_1 > 1$ ensures finite expected stock price. By choosing $q > p$ and calibrating η_2 independently of η_1, the model captures the empirically observed asymmetry between large negative and positive moves — something the symmetric normal jumps in Merton (1976) cannot reproduce.

Why Double Exponential: Intuition and Tractability

The double exponential distribution is motivated on two grounds. Economically, it links to behavioral finance: the heavy tails model investor *overreaction* to news (large jumps), while the high peak models *underreaction* (small price responses). The asymmetry between upward and downward components directly explains the volatility skew, since higher probability of large downward jumps inflates implied volatility for low-strike puts.

Mathematically, the key advantage is the *memoryless property* of the exponential distribution. When a jump process crosses a barrier, it often overshoots. Computing the exact distribution of this overshoot is generally intractable, but for exponential jump sizes the memoryless property gives a closed-form overshoot distribution and ensures the overshoot is conditionally independent of the first passage time. This is precisely why the Kou model yields analytical solutions for barrier, lookback, and perpetual American options, while the Merton model with normal jumps cannot.

For European options, the pricing formula uses the Hh function:

$$Hh_n(x) = \frac{1}{n!}\int_x^\infty (t-x)^n e^{-t^2/2}\,dt, \qquad n = 0, 1, 2, \ldots \tag{3}$$

which satisfies a simple three-term recursion and is computed efficiently from the standard normal distribution. In practice, truncating the series to ten to fifteen terms gives accurate results.

Strengths, Limitations, and Importance

The model's principal strengths are its analytical tractability across vanilla and path-dependent derivatives, its asymmetric jump structure that fits the volatility skew, and the clear economic interpretation of all parameters (λ is mean jumps per year; p, q are jump direction probabilities; $1/\eta_i$ are mean jump sizes). It improves on Merton (1976) both empirically and mathematically, and it can be embedded in a rational expectations equilibrium, making its pricing formula preference-free.

The main limitations mirror those of the Merton model. The constant jump intensity λ cannot capture time-varying crash risk, and the Poisson independence assumption rules out volatility clustering. The constant diffusion volatility σ ignores the well-documented mean-reversion of return volatility; models combining stochastic volatility with jumps, such as Bates (1996), address this. The market remains incomplete under jump dynamics, so additional equilibrium assumptions are needed to uniquely pin down prices.

Relevance to QQQ Options

The QQQ ETF tracks the NASDAQ-100, a technology-heavy index prone to large, sudden price moves driven by earnings surprises, regulatory decisions, and product announcements. The QQQ implied volatility surface shows a pronounced downside skew, with out-of-the-money puts carrying materially higher implied volatilities than at-the-money options. The Kou model's asymmetric jump structure is well suited to match this pattern by assigning higher probability and larger size to downward jumps through the parameters q and η_2. We calibrate $(\sigma, \lambda, p, \eta_1, \eta_2)$ to the QQQ option cross-section and compare the fit against Black–Scholes and Merton. The Kou model is expected to outperform both benchmarks for short-maturity, far-from-the-money options, especially on the put side, while converging toward the Merton fit for longer-dated near-the-money contracts where the diffusion component dominates.